Fan Ding

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EDUCATION

Michigan Technological University — **Houghton, Michigan**
Ph.D. *in Computer Science* — *2025*

Hebei University of Technology — **Tianjin, China**
B.Eng. *in Software Engineering* — *2017*

WORK EXPERIENCE

Michigan Technological University — **Houghton, Michigan**
PhD Researcher — *2018 – 2026*

- User Research & Experimentation: Led user studies for Virtual Reality (VR) and acoustic systems; managed data collection protocols and engineered laboratory environments integrating Oculus Head-Mounted Displays (HMDs) and Vicon infrared tracking systems.
- VR Spatial Perception: Developed real-time interactive 3D environments in C++ (OpenGL) and analyzed human-subject data to quantify perceptual biases, establishing empirical guidelines for VR display design.
- Acoustic Surface Interaction: Built a microphone array system for touch detection, developing a robust 8×96kHz signal processing pipeline in Python with a modular design and client-server architecture to enable alternative input methods in Augmented Reality (AR).

National Laboratory of Pattern Recognition
Institute of Automation, Chinese Academy of Sciences (CAS) — **Beijing, China**
Research Assistant — *2016 – 2018*

- Developed C++ geometry processing algorithms for mesh optimization and user-guided segmentation, implementing plugins for the Graphite 3D modeling platform.
- Designed and evaluated algorithms for robust Computer-Aided Design (CAD) tessellation using both public benchmarks and real industrial CAD models from industry partners.
- Collaborated on “Geometric Feature Recognition Technology for Simulation of Complete Appliance” program in partnership with Huawei Technologies, including implementation, testing, and validation.
- Supported publication of peer-reviewed research in IEEE TVCG and Computational Visual Media.

TEACHING EXPERIENCE

Michigan Technological University — **Houghton, Michigan**
Teaching Assistant — *2018 – 2025*

- Courses: Computer Graphics, Systems Programming, Database Systems, Discrete Structures
- Supported undergraduate instruction through office hours, debugging support, and project guidance.

SELECTED PUBLICATIONS

Ding, F., Sepahyar, S., & Kuhl, S. Effects of Brightness on Distance Judgments in Head-Mounted Displays. *ACM Symposium on Applied Perception (SAP),* 2020.

Guo, J., Jia, X., Yan, D.-M., & Ding, F. Automatic and High-Quality Surface Mesh Generation for CAD Models. *Computer-Aided Design,* 2019.

Khan, D., Yan, D.-M., Ding, F., & Zhang, X. Surface Remeshing with Robust User-Guided Segmentation. *Computational Visual Media (Springer),* 2018.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, HTML5, CSS3, JavaScript, SQL

AI & Machine Learning: Machine Learning, Large Language Models, Prompt Engineering, Agent Workflows

Graphics & XR: Unity3D, OpenGL, WebGL, Oculus SDK, Microsoft HoloLens, MRTK, Vicon, Blender

Area of Expertise: CAD Geometry/Mesh Processing, Virtual Environments (VR/AR), Acoustic Signal Processing, Spatial Interaction, Human-Computer Interaction (HCI)

Tools & Libraries: Git, Linux, NumPy, SciPy, Three.js, Matplotlib